USES OF SYNTHETIC COMPOSITES

Breathable fabric
Traditional waterproof clothing traps sweat inside. Composite versions mix nylon with polytetrafluoroethylene (PTFE), which does not allow rainwater to pass through but lets water molecules from sweat escape.

Disc brakes
Some high-performance cars and heavy vehicles use disc brakes made from carbon-fiber-reinforced ceramics. Not only is the material lightweight and strong, but it also has exceptionally high heat tolerance.

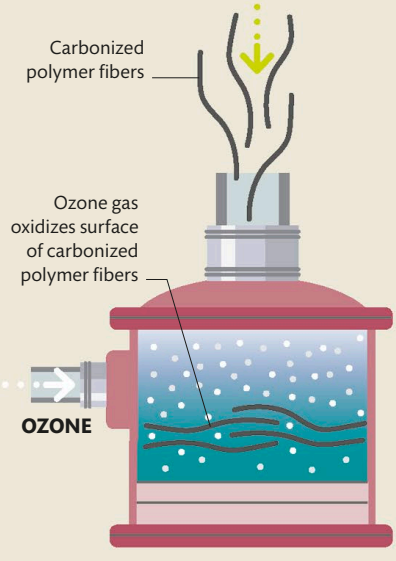
Bicycle frames
The frames of most racing bicycles are made from a variety of carbon fibers of different types, each used in different places for very specific purposes. Carbon fiber is also used to make some other components, such as wheels and handlebars.

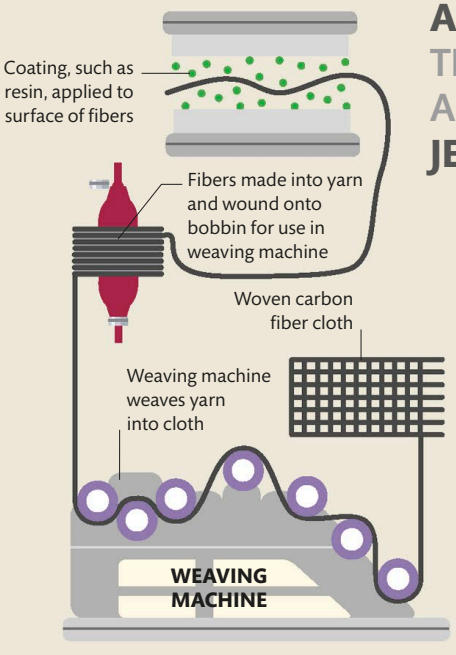
Boat hulls
Fiberglass has been widely used in boat-hull construction since the 1950s. At the leading edge of boat building, composites using aramid fibers—a particularly strong fiber used in aerospace—are used to reinforce key areas.

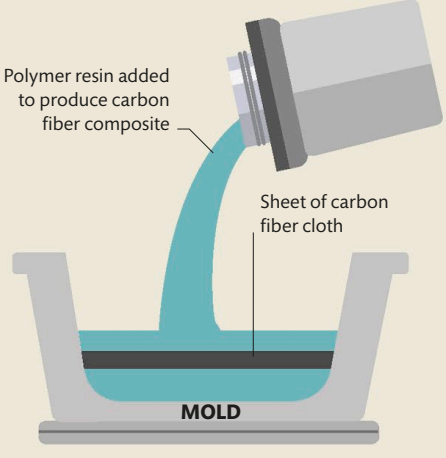
Kevlar
Kevlar is a composite fiber that is about five times stronger than steel. It can be woven into cloth to create bulletproof garments, used to create mooring lines, or added to polymers to form racing sails or linings for bicycle tires.

Reinforced concrete
One of the oldest and most common synthetic composites, concrete is a mix of cement, water, sand, and gravel (see pp.76–77). Its poor tensile strength can be improved by embedding steel bars within the concrete.

AROUND HALF OF THE MATERIALS IN A MODERN PASSENGER JET ARE COMPOSITES







4 Surface of fibers oxidized
After carbonizing, the fibers have a surface that does not bond well. To improve the bonding properties, their surface is slightly oxidized by the addition of oxygen atoms from ozone.

5 Fibers coated and woven
After the surface treatment, the fibers are coated for protection and twisted together to form yarn. The yarn is wound onto bobbins, which are loaded onto weaving machines to produce cloth.

6 Carbon fiber polymer produced
The carbon fiber cloth is delivered to manufacturers to complete the process to their individual requirements. This involves placing it into a mold and adding polymer resin to form the composite.